

OTTAWA GATINEAU, QC, Canada

WMO: 716279

Lat: 45.520N

Lon: 75.563W

Elev: 64

StdP: 100.55

Time Zone: -5.00 (NAE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-24.5	-31.7	0.2	-26.6	-29.3	0.3	-23.9	10.4	-5.4	9.3	-8.5	1.3	0	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	31.1	22.0	29.5	21.4	27.9	20.5	23.8	28.6	22.8	27.3	21.8	26.3	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.1	26.5	21.3	16.1	25.3	20.3	15.1	24.2	71.7	28.7	67.6	27.4	64.1	26.3	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.0	7.0	-32.3	33.8	1.7	0.9	-33.5	34.4	-34.4	34.9	-35.4	35.4	-36.6	36.1		
			-32.4	25.8	1.6	1.3	-33.6	26.8	-34.5	27.5	-35.5	28.2	-36.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-10.0	-9.5	-4.0	5.0	13.2	17.4	20.8	19.6	14.7	7.5	0.5	-7.0
	DBStd	12.23	7.28	6.76	6.15	4.79	4.47	3.73	2.82	3.18	4.56	4.43	5.39	7.61
	HDD10.0	2709	619	547	434	161	18	1	0	0	10	103	288	528
	HDD18.3	4795	877	780	692	399	169	61	9	24	126	334	536	786
	CDD10.0	1164	0	0	0	13	118	222	335	296	152	27	2	0
	CDD18.3	207	0	0	0	1	9	32	86	62	17	0	0	0
	CDH23.3	2201	0	0	0	16	158	404	909	539	171	4	0	0
	CDH26.7	621	0	0	0	5	34	119	291	138	34	0	0	0
Wind	WSAvg	3.0	3.3	3.3	3.4	3.6	3.1	2.8	2.3	2.2	2.3	2.8	3.3	3.2
Precipitation	PrecAvg	907	59	59	64	74	77	89	85	82	86	80	87	77
	PrecMax	1116	97	139	130	132	160	177	124	204	140	147	154	118
	PrecMin	722	12	2	15	30	31	31	36	11	43	27	42	23
	PrecStd	106	24	31	33	28	35	39	26	41	32	33	29	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	6.8	11.1	25.1	29.9	32.5	33.9	31.9	30.1	23.3	16.6	8.7
		MCWB	4.1	4.9	7.6	18.3	19.1	21.4	23.7	23.2	23.4	18.1	13.3	7.0
	2%	DB	3.7	4.2	7.3	19.7	27.2	29.9	31.7	29.7	27.3	20.3	12.8	6.0
		MCWB	2.1	2.1	4.2	12.1	18.3	21.1	22.3	22.1	21.3	16.0	10.6	4.5
	5%	DB	2.0	2.8	5.3	15.9	24.9	27.7	29.9	28.2	25.4	17.5	10.5	3.9
		MCWB	0.8	1.3	2.8	10.3	16.8	19.4	21.6	21.4	20.0	13.8	8.6	2.7
	10%	DB	0.6	1.3	3.7	13.0	22.5	25.5	28.0	26.5	23.0	14.9	7.9	2.3
		MCWB	-0.3	0.3	1.8	8.2	15.6	18.4	20.4	20.6	18.8	11.8	6.1	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	4.9	9.0	18.6	22.0	24.2	25.6	24.7	24.2	19.3	13.7	7.5
		MCDB	5.7	6.1	11.2	25.8	26.8	28.8	31.4	29.6	28.4	21.6	15.9	8.7
	2%	WB	2.3	2.7	4.9	13.7	20.1	22.4	23.8	23.4	22.5	16.8	11.1	4.7
		MCDB	3.2	3.8	6.3	17.3	25.1	27.6	29.1	27.9	26.1	19.0	12.3	5.9
	5%	WB	1.0	1.5	3.4	11.0	18.1	21.0	22.8	22.4	21.1	14.4	8.5	2.9
		MCDB	1.9	2.5	4.7	15.7	22.9	25.9	27.8	26.6	23.9	16.9	10.1	3.8
	10%	WB	-0.2	0.3	2.1	9.0	16.4	19.5	21.7	21.3	19.5	12.2	6.3	1.3
		MCDB	0.7	1.5	3.5	12.2	20.5	23.7	26.6	25.0	22.5	14.1	8.0	2.1
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.6	10.1	10.8	13.0	12.5	13.1	11.8	12.4	10.5	8.4	8.8
		MCDBR	9.9	11.0	10.0	16.8	17.1	16.0	15.5	13.9	14.4	14.4	11.8	8.7
		MCWBR	9.4	9.9	7.9	10.3	9.5	8.2	7.6	7.4	8.8	10.1	9.9	7.9
	5% WB	MCDBR	9.1	10.5	9.1	14.4	14.1	12.6	13.3	11.8	13.1	13.1	11.5	8.4
		MCWBR	8.9	9.8	7.8	9.4	8.8	7.4	7.4	6.8	8.8	10.1	10.1	8.0
Clear-Sky Solar Irradiance	taub	0.261	0.287	0.327	0.345	0.384	0.415	0.419	0.400	0.371	0.339	0.310	0.278	
	taud	2.365	2.283	2.322	2.412	2.347	2.273	2.269	2.327	2.398	2.483	2.489	2.378	
	Ebn at Noon	859	893	902	915	885	855	846	849	846	828	794	792	
	Edh at Noon	73	97	109	109	121	131	130	118	101	81	67	66	
All-Sky Solar Radiation	RadAvg	1.40	2.38	3.70	4.50	5.18	5.64	5.74	5.06	3.95	2.33	1.42	1.01	
	RadStd	0.17	0.23	0.32	0.45	0.50	0.43	0.47	0.24	0.32	0.20	0.17	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page